

Chapter 3, Section 4

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1. Let $f : X \rightarrow Y$ be an affine morphism of Noetherian separated schemes (II, Ex. 5.17). Show that for any quasi-coherent sheaf $\mathcal{F}$ on X , there are natural isomorphisms for all $i \geq 0$,

$$H^i(X, \mathcal{F}) \cong H^i(Y, f_*\mathcal{F}).$$

Proof. Let $\mathfrak{V}$ be an open affine cover of Y so that $\mathfrak{U} = (f^{-1}(V))_{V \in \mathfrak{V}}$ is an open affine cover of X . We can compute $H^i(X, \mathcal{F})$ using the Čech complex defined by the open covering $\mathfrak{U}$. Also, $f_*\mathcal{F}$ is a quasi-coherent $\mathcal{O}_Y$ -module by (II, 5.8), so the cohomology of $f_*\mathcal{F}$ can be computed via the Čech complex defined by $\mathfrak{V}$ (4.5). Lastly, $f_*\mathcal{F}(V) = \mathcal{F}(U)$ for all $V \in \mathfrak{V}$ and corresponding $U = f^{-1}(V) \in \mathfrak{U}$, so the Čech cohomology of $\mathcal{F}$ and $f_*\mathcal{F}$ with respect to $\mathfrak{U}$ and $\mathfrak{V}$, respectively, are isomorphic. Hence, $H^i(X, \mathcal{F}) \cong H^i(Y, f_*\mathcal{F})$ for all $i \geq 0$ by (4.5). $\square$

2. Prove Chevalley's theorem: Let $f : X \rightarrow Y$ be a finite surjective morphism of Noetherian separated schemes, with X affine. Then Y is affine.
 - (a) Let $f : X \rightarrow Y$ be a finite surjective morphism of integral Noetherian schemes. Show that there is a coherent sheaf $\mathcal{M}$ on X , and a morphism of sheaves $\alpha : \mathcal{O}_Y^r \rightarrow f_*\mathcal{M}$ for some $r > 0$, such that α is an isomorphism at the generic point of Y .
 - (b) For any coherent sheaf $\mathcal{F}$ on Y , show that there is a coherent sheaf $\mathcal{G}$ on X , and a morphism $\beta : f_*\mathcal{G} \rightarrow \mathcal{F}^r$ which is an isomorphism at the generic point of Y .
 - (c) Now prove Chevalley's theorem.

Proof.

- (a) Let $\mathcal{L}$ and $\mathcal{K}$ be the sheaf of total quotient rings of X and Y , respectively. Since the function field of X is a finite field extension of that of Y , there exists an isomorphism $\phi : \mathcal{K}^r \rightarrow f_*\mathcal{L}$ for some $r > 0$. Let $\alpha : \mathcal{O}_Y^r \rightarrow f_*\mathcal{L}$ be the composition of the natural map $\mathcal{O}_Y^r \rightarrow \mathcal{K}^r$ with ϕ . This is an isomorphism at the generic point of Y . Replacing $\mathcal{L}$ by the sub- $\mathcal{O}_X$ -module spanned by the image of $\mathcal{K}^r$, we obtain the desired $\mathcal{M}$.
- (b) Let $\mathcal{F}$ be any coherent sheaf $\mathcal{F}$ on Y , $\mathcal{A} = f_*\mathcal{O}_X$, and $\alpha : \mathcal{O}_Y^r \rightarrow f_*\mathcal{M}$ as in (a) for some coherent sheaf $\mathcal{M}$ on X . Applying $\mathbf{Hom}(\cdot, \mathcal{F})$ to α gives $\beta : \mathbf{Hom}(f_*\mathcal{M}, \mathcal{F}) \rightarrow \mathcal{F}^r$, which is an isomorphism at the generic point of Y . The $\mathcal{O}_Y$ -module $\mathbf{Hom}(f_*\mathcal{M}, \mathcal{F})$ is naturally a coherent $\mathcal{A}$ -module, and finite morphisms are also affine, so by the equivalence of categories of coherent $\mathcal{O}_X$ -modules and coherent $\mathcal{A}$ -modules (II, 5.17e), there exists a coherent sheaf $\mathcal{G}$ on X such that $f_*\mathcal{G} \cong \mathbf{Hom}(f_*\mathcal{M}, \mathcal{F})$.
- (c) Let X_{red} and Y_{red} be the reduced schemes of X and Y , respectively (II, Ex. 2.3). By the universal property of $Y_{\text{red}} \rightarrow Y$, there exists a unique morphism $X_{\text{red}} \rightarrow Y_{\text{red}}$ making the following diagram commute

$$\begin{array}{ccc} X_{\text{red}} & \longrightarrow & Y_{\text{red}} \\ \downarrow & & \downarrow \\ X & \longrightarrow & Y \end{array}$$

A scheme is affine if and only if its reduced scheme is affine (Ex. 3.1) so assume X and Y are reduced by replacing $X \rightarrow Y$ with $X_{\text{red}} \rightarrow Y_{\text{red}}$. Because f is surjective, every irreducible component of Y has a nonempty irreducible preimage and has a structure of a closed subscheme of X . All closed subschemes of X are affine (II, Ex. 3.11b), so we reduce to the case X and Y integral.

Let $\mathcal{F}$ be a coherent sheaf on Y . We will show $H^i(Y, \mathcal{F}^r) = 0$ for all $i > 0$ (2.9) (3.7). Let $\beta : f_*\mathcal{G} \rightarrow \mathcal{F}^r$ as in (b). Suppose the set Σ of proper closed subsets of Y which are not affine is nonempty, and let $Z \subset Y$ be a minimal element in Σ . Thus, all proper closed subsets of Z are affine. By replacing X and Y with $f^{-1}Z$ and Z , and $\mathcal{F}$ with any coherent sheaf $\mathcal{F}$ with support in Z , we show Z must be affine, i.e., assume every proper closed subset of Y is affine, and by noetherian induction, this implies Y itself is affine.

By (II, 5.7), $\ker \beta$ and $\text{coker } \beta$ are coherent sheaves on Y , so $K = \text{Supp}(\ker \beta)$ and $C = \text{Supp}(\text{coker } \beta)$ are closed subsets of Y . Also, K and C are both proper subsets because β is an isomorphism at the generic point, so by hypothesis, K and C are affine. For any closed subset $Z \subseteq Y$, let $j : Z \rightarrow Y$ denote the inclusion map. We have $\beta \cong j_* j^* \ker \beta$ and similarly for $\text{coker } \beta$, so by (3.7) and (Ex. 4.1),

$$H^i(Y, \ker \beta) \cong H^i(K, j^* \ker \beta) = 0, \quad H^i(Y, \text{coker } \beta) \cong H^i(C, j^* \text{coker } \beta) = 0.$$

The exact sequence

$$0 \longrightarrow \ker \beta \longrightarrow f_*\mathcal{G} \longrightarrow \mathcal{F}^r \longrightarrow \text{coker } \beta \longrightarrow 0$$

induces two long exact sequences in cohomology

$$0 \longrightarrow H^0(Y, \ker \beta) \longrightarrow H^0(Y, f_* \mathcal{G}) \longrightarrow H^0(Y, \operatorname{im} \beta) \longrightarrow \dots,$$

$$0 \longrightarrow H^0(Y, \operatorname{im} \beta) \longrightarrow H^0(Y, \mathcal{F}^r) \longrightarrow H^0(Y, \operatorname{coker} \beta) \longrightarrow \dots.$$

Again by (Ex. 4.1), $H^i(Y, f_* \mathcal{G}) \cong H^i(X, \mathcal{G}) = 0$. By exactness of above, we conclude that $H^i(Y, \mathcal{F}^r) = 0$.

□

3. Let $X = \mathbb{A}_k^2 = \operatorname{Spec} k[x, y]$, and let $U = X - \{(0, 0)\}$. Using a suitable cover of U by open affine subsets, show that $H^1(U, \mathcal{O}_U)$ is isomorphic to the k -vector space spanned by $\{x^i y^j \mid i, j < 0\}$. In particular, it is infinite-dimensional.

Proof. Let $\mathcal{U}$ be the open covering by the two open sets $V = X - \{x = 0\}$ and $U = X - \{y = 0\}$, with affine coordinates obtained by restricting the ones from X . Then the Čech complex has only two terms:

$$C^0 = \Gamma(V, \mathcal{O}_V) \times \Gamma(W, \mathcal{O}_W),$$

$$C^1 = \Gamma(V \cap W, \mathcal{O}_{V \cap W}).$$

Now

$$\Gamma(V, \mathcal{O}) = k \left[x, \frac{1}{x}, y \right]$$

$$\Gamma(W, \mathcal{O}) = k \left[x, y, \frac{1}{y} \right]$$

$$\Gamma(V \cap W, \mathcal{O}) = k \left[x, y, \frac{1}{x}, \frac{1}{y} \right]$$

and the map $d : C^0 \rightarrow C^1$ is given by $(f, g) \mapsto f - g$. To compute H^1 , the image of d is the set of all expressions $x^i y^j$ where at least one of i, j is non-negative. Hence, $H^1(U, \mathcal{O}_U)$ is spanned by $\{x^i y^j \mid i, j < 0\}$. □

4. On an arbitrary topological space X with an arbitrary Abelian sheaf $\mathcal{F}$, Čech cohomology may not give the same result as the derived functor cohomology. But here we show that for H^1 , there is an isomorphism if one takes the limit over all coverings.

- (a) Let $\mathcal{U} = (U_i)_{i \in I}$ be an open covering of the topological space X . A *refinement* of $\mathcal{U}$ is a covering $\mathcal{V} = (V_j)_{j \in J}$, together with a map $\lambda : J \rightarrow I$ of the index sets, such that for each $j \in J$, $V_j \subseteq U_{\lambda(j)}$. If $\mathcal{V}$ is a refinement of $\mathcal{U}$, show that there is a natural induced map on Čech cohomology for any Abelian sheaf $\mathcal{F}$, and for each i ,

$$\lambda^i : \check{H}^i(\mathcal{U}, \mathcal{F}) \rightarrow \check{H}^i(\mathcal{V}, \mathcal{F}).$$

The coverings of X form a partially ordered set under refinement, so we can consider the Čech cohomology in the limit

$$\varinjlim_{\mathcal{U}} \check{H}^i(\mathcal{U}, \mathcal{F}).$$

- (b) For any Abelian sheaf $\mathcal{F}$ on X , show that the natural maps (4.4) for each covering

$$\check{H}^i(\mathcal{U}, \mathcal{F}) \rightarrow H^i(X, \mathcal{F})$$

are compatible with the refinement maps above.

- (c) Now prove the following theorem. Let X be a topological space, $\mathcal{F}$ a sheaf of Abelian groups. Then the natural map

$$\varinjlim_{\mathcal{U}} \check{H}^1(\mathcal{U}, \mathcal{F}) \rightarrow H^1(X, \mathcal{F})$$

is an isomorphism.

Proof.

- (a) An element $\alpha \in C^p(\mathcal{U}, \mathcal{F})$ is determined by giving an element

$$\alpha_{i_0, \dots, i_p} \in \mathcal{F}(U_{i_0, \dots, i_p}),$$

for each $(p+1)$ -tuple $i_0 < \dots < i_p$ of elements of I . We define the map $\Lambda^p : C^p(\mathcal{U}) \rightarrow C^p(\mathcal{V})$ by setting

$$(\Lambda \alpha)_{j_0, \dots, j_p} = \alpha_{\lambda(j_0), \dots, \lambda(j_p)} \Big|_{V_{j_0, \dots, j_p}}$$

if $\lambda(j_i)$ are all distinct, and 0 otherwise. One has $d_{\mathcal{V}} \Lambda = \Lambda d_{\mathcal{U}}$, so Λ induces a map of cohomology groups.

(b) We want to show the following diagram commutes

$$\begin{array}{ccc} & \mathcal{J}^* & \\ \nearrow & & \nwarrow \\ \mathcal{C}^*(\mathcal{U}, \mathcal{F}) & \longrightarrow & \mathcal{C}^*(\mathcal{V}, \mathcal{F}) \end{array}$$

where $0 \rightarrow \mathcal{F} \rightarrow \mathcal{J}^*$ is any injective resolution of $\mathcal{F}$ in $\mathcal{A}b(X)$. This follows from the universal property of $\mathcal{J}^*$. In particular, there is only one extension of $\mathcal{F} \rightarrow \mathcal{J}^*$ to $\mathcal{C}^*(\mathcal{U}, \mathcal{F})$.

(c) Embed $\mathcal{F}$ in a flasque sheaf $\mathcal{G}$, and let $\mathcal{R} = \mathcal{G}/\mathcal{F}$, so that we have an exact sequence

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{G} \longrightarrow \mathcal{R} \longrightarrow 0.$$

Define a complex D^* by

$$0 \longrightarrow C^*(\mathcal{U}, \mathcal{F}) \longrightarrow C^*(\mathcal{U}, \mathcal{G}) \longrightarrow D^*(\mathcal{U}) \longrightarrow 0.$$

For any refinement $\mathcal{V}$ of $\mathcal{U}$, we have a commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & C^*(\mathcal{U}, \mathcal{F}) & \longrightarrow & C^*(\mathcal{U}, \mathcal{G}) & \longrightarrow & D^*(\mathcal{U}) \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & C^*(\mathcal{V}, \mathcal{F}) & \longrightarrow & C^*(\mathcal{V}, \mathcal{G}) & \longrightarrow & D^*(\mathcal{V}) \longrightarrow 0 \end{array}$$

Let $H_D^i(\mathcal{U})$ be the cohomology of the complex $D^*(\mathcal{U})$. The natural map of complex

$$D^*(\mathcal{U}) \rightarrow C^*(\mathcal{U}, \mathcal{R})$$

induces a natural map of cohomology groups $H_D^i(\mathcal{U}) \rightarrow \check{H}^i(\mathcal{U}, \mathcal{R})$. Noting that the Čech cohomology of $\mathcal{G}$ vanishes for $i > 0$, consider the following commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & \varinjlim_{\mathcal{U}} \check{H}^0(\mathcal{U}, \mathcal{F}) & \longrightarrow & \varinjlim_{\mathcal{U}} \check{H}^0(\mathcal{U}, \mathcal{G}) & \longrightarrow & \varinjlim_{\mathcal{U}} H_D^0(\mathcal{U}) \longrightarrow \varinjlim_{\mathcal{U}} \check{H}^1(\mathcal{U}, \mathcal{F}) \longrightarrow 0 \\ & & \downarrow \wr & & \downarrow \wr & & \downarrow \\ 0 & \longrightarrow & H^0(X, \mathcal{F}) & \longrightarrow & H^0(X, \mathcal{G}) & \longrightarrow & H^0(X, \mathcal{R}) \longrightarrow H^1(X, \mathcal{F}) \longrightarrow 0 \end{array}$$

By the Snake lemma, it suffices to show

$$\varinjlim_{\mathcal{U}} H_D^0(\mathcal{U}) \cong H^0(X, \mathcal{R}).$$

□

5. For any ringed space $(X, \mathcal{O}_X)$, let $\text{Pic } X$ be the group of isomorphism classes of invertible sheaves (II, §6). Show that $\text{Pic } X \cong H^1(X, \mathcal{O}_X^*)$ where $\mathcal{O}_X^*$ denotes the sheaf whose sections over an open set U are the units in the ring $\Gamma(U, \mathcal{O}_X)$, with multiplication as the group operation.

Proof. For any invertible sheaf $\mathcal{L}$ on X , let $\mathcal{U} = (U_i)_{i \in I}$ be an open cover of X such that $\mathcal{L}|_{U_i}$ is free for each i . Fix isomorphisms $\varphi_i : \mathcal{O}_{U_i} \rightarrow \mathcal{L}|_{U_i}$. Then on $U_i \cap U_j$, we get an isomorphism $\varphi_i^{-1} \circ \varphi_j$ of $\mathcal{O}_{U_i \cap U_j}$ with itself. In particular, $(\varphi_i^{-1} \circ \varphi_j)(1)$ is a unit in $\mathcal{O}_{U_i \cap U_j}$. For another $U_k \in \mathcal{U}$, one has

$$(\varphi_j^{-1} \circ \varphi_k)(1) \Big|_{U_i \cap U_j \cap U_k} \cdot (\varphi_i^{-1} \circ \varphi_j)(1) \Big|_{U_i \cap U_j \cap U_k} = (\varphi_i^{-1} \circ \varphi_k)(1) \Big|_{U_i \cap U_j \cap U_k}$$

by (I, Ex. 1.22), so the isomorphisms $\varphi_i^{-1} \circ \varphi_j$ give an element of $\check{H}^1(\mathcal{U}, \mathcal{O}_X^*)$. Define a map

$$L : \varinjlim_{\mathcal{U}} \check{H}^1(\mathcal{U}, \mathcal{O}_X^*) \rightarrow \text{Pic } X$$

as follows: for every $\alpha \in \check{H}^1(\mathcal{U}, \mathcal{O}_X^*)$, define $L(\alpha)$ to be the invertible sheaf on X obtained by glueing $\mathcal{O}_{U_i}$ via the transition functions defined by $\varphi_{ij}(1) = \alpha_{ij}$. This is an isomorphism by the previous remarks, hence $\text{Pic } X \cong H^1(X, \mathcal{O}_X^*)$ by (Ex. 4.4). □